

EXP-9

CODE:

```
clc;
clear;
close all;

%% Fixed Input Sequence
input_bits = [1 1 0 1 0 0];

%% Convolutional Encoder
trellis = poly2trellis(3,[7 5]);
encoded_bits = convenc(input_bits, trellis);

%% Separate Encoded Bits into Pairs
encoded_pairs = reshape(encoded_bits, 2, []);

%% Create Output Strings
output_pair = cell(length(input_bits),1);
for i = 1:length(input_bits)
    output_pair{i} = sprintf('%d%d', ...
        encoded_pairs(i,1), encoded_pairs(i,2));
end

%% Create Table
Bit_Position = (1:length(input_bits))';
Input_Bit = input_bits';
Output_Encoded = output_pair;
T = table(Bit_Position, Input_Bit, Output_Encoded);
disp('Convolutional Encoding Table');
disp(T);
```

```
%% Plot Input and Encoded Bit Sequences

figure;
subplot(2,1,1);
stairs(0:length(input_bits)-1, input_bits, 'LineWidth', 1.5);
ylim([-0.2 1.2]);
title('Input Bit Sequence');
xlabel('Bit Position');
ylabel('Bit Value');
grid on;
subplot(2,1,2);
stairs(0:length(encoded_bits)-1, encoded_bits, 'LineWidth', 1.5);
ylim([-0.2 1.2]);
title('Encoded Output Bit Sequence');
xlabel('Bit Position');
ylabel('Bit Value');
grid on;

%% Display Results
fprintf('\nInput Bits : ');
fprintf('%d ', input_bits);
fprintf('\nEncoded Bits : ');
fprintf('%d ', encoded_bits);

fprintf('\n');
```

OUTPUT:

Figure 1 ×

